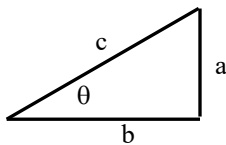


If this were a quiz, it would be closed book, and closed notes. Calculators are permitted only to do mathematical calculations. The **ONLY** physics formulas that may be used are those given below. Define all symbols and justify all mathematical expressions used. Make sure to state all assumptions & approximations used to solve a problem. Partial credit would be given for a well-communicated problem-solving framework based on correct physics. On a quiz, each problem is worth 25 points: **In the context of a unified solution**, credit will be awarded as follows: a useful sketch that shows all relevant quantities, listing the known quantities, and writing the quantity to be determined is 7 points; writing the physics principles useful to solve the problem, writing any assumptions or approximations needed, drawing useful physics diagrams, and writing the equations that apply the physics principles in terms of the quantities from your sketch is 6 points; planning the solution by describing how to use those physics equations to construct an algebraic answer is 5 points; solving those equations for a symbolic result is 3 points; checking the units & reasonableness of the symbolic result is 2 points; calculating a numerical result & stating why it units are correct and giving a reason why that result is not unreasonable is 2 points. If no numerical answer is required, 1 point will be added the planning the solution and 1 point to solving the equations and checking the units & reasonableness of the solution.

Useful Mathematical Relationships:



For a right triangle: $\sin \theta = \frac{a}{c}$, $\cos \theta = \frac{b}{c}$, $\tan \theta = \frac{a}{b}$,

$$a^2 + b^2 = c^2, \sin^2 \theta + \cos^2 \theta = 1$$

For a circle: $C = 2\pi R$, $A = \pi R^2$

For a sphere: $V = \frac{4}{3}\pi R^3$, $S = 4\pi R^2$

If $Ax^2 + Bx + C = 0$, $x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$, $(a + b)^2 = a^2 + 2ab + b^2$

Fundamental Concepts, Principles, and Definitions:

$\sum \vec{F} = m\vec{a}$	$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$	$a_{x \text{ average}} = \frac{\Delta v_x}{\Delta t}$	$v_{x \text{ average}} = \frac{\Delta x}{\Delta t}$
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Under Certain Conditions:

$F = mg$	$F = \mu N$	$F_{\max} = \mu N$
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Useful constants: 1 mile = 5280 ft, $g = 9.8 \text{ m/s}^2 = 32 \text{ ft/s}^2$

While watching a TV show about ancient architecture, you wonder how civilizations several thousand years ago could move heavy stone blocks along the top of the walls that surround their cities. One theory is that they used a rope to pull a heavy block horizontally along the flat top of a high wall that is covered in grease by attaching it to a rope whose other end was attached to a much lighter block that hung straight down over the edge of the wall. You decide to test this idea by seeing how well you could move a heavy block on a flat surface with a light block hanging straight down over the edge of that surface by calculating how the acceleration of the heavy block depends on the frictional force on it and the weight of both the heavy and light block. You imagine that the horizontal rope from the heavy block runs over a rail on a platform at the edge of the wall that allows the rope holding the lighter block to hang straight down. The platform rail that the rope runs over is covered in grease so that it is very slippery.